Practical Lab: Access Control Models

El Gourji Nasreddine

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1 Challenge 1 :

1.1 Tasks:

1.1.1 Answer to the question : Suppose the directory has permission 750, but a file inside it has permission 600. Can a member of the accounting group read the file ? Justify your answer.

No , the Accounting group won't be able to read the file because 750 of the parent directory means they can only list the reading of the file and the access only depends on wether the file itself permitted or not , and since we have 0 for the members of the accounting group this means they can't read it .

2 Challenge 2 :

2.1 Tasks:

2.1.1 Constructing the access matrix indicating which documents each user can read.

User	Secret	Confidential	Public
Alice	Yes	Yes	Yes
Bob	No	Yes	Yes
Charlie	No	No	Yes

2.1.2 Bob attempts to read Document_A. Should access be granted ?

No , since the Document_A is Secret and using the matrix above Bob shouldn't be allowed to according to the No read up.

2.1.3 Alice attempts to write information into Document_C. Is this operation allowed ?

No , since the No write down policy don't let anyone from the upper clearence write what's below .

2.1.4 Explain how Bell-LaPadula prevents information leakage.

With No Read up and No Write Down , secret information can never end up somewhere a lower clearance user could read it.

2.1.5 A user has sufficient clearance to read a document, but the document owner explicitly denies access using ordinary Linux permissions. Is access necessarily granted under MAC ? Explain the difference between DAC and MAC enforcement.

No , since they should both agree on giving the privielege if one isn't satisfied you don't get the access . DAC is about clear defined rules by the owners while the MAC is about the hierachy .

3 Challenge 3 :

3.1 Tasks :

3.1.1 3. Identify whether the previous configuration satisfies the complete access policy.

1. for the source : Administrators aren't configurated at all no bit can specify for them so they don't have any access
2. For the reports : Administrators also have no bit left specificcally for them , so they have no access

3.1.2 Correct any missing permissions.

a simple fix would be to add charlie to Viewers and developers additionally to administrators .but we still gonna miss thhe Full access for the reports

4 Challenge 4 :

4.1 Tasks

4.1.1 Determine whether each request is granted or denied.

User	Decision
Alice	Granted
Bob	Denied
Charlie	Denied
David	Denied

4.1.2 Identify the attribute responsible for each denial.

1. Bob : Document != Department
2. Charlie : Public ¡ Secret
3. David : 22:00 ¿ 18:00

4.1.3 Alice requests access to an HR document. What happens ?

She gets denied , beacuse her departement won't match the document .

4.1.4 David requests access at 17 :00. Does the decision change

Yes , now he will be granted because 17:00 ¡ 18:00

4.1.5 Explain why ABAC is more flexible than RBAC.

RBAC only relies on role , while ABAC takes into consideration more parameters and must all be satisfied which make it more flexible and easily modifiable .